Homework 12 - Multiple Regression

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Biostat Assignment: Multiple Regression

Chapter 19, Problem 7

Variable	Coefficient	Standard Error
Diet Group	-11.25	4.33
Baseline Cholesterol	0.85	0.07
Body Mass Index	0.23	0.65
Gender	-3.02	4.42

a. Conduct tests of the null hypothesis that each of the four coefficients in the regression equation is equal to 0. At the 0.05 level of significance, which of the explanatory variables have an effect on serum cholesterol level eight weeks after the start of the study?

```
library(dplyr)
library(tidyr)
library(gt)

#| label: coef-table
# With unknown residuals, the normal approximation (p_norm) is fine for large samples.
df_resid <- 100

coef_df <- tibble::tribble(
  ~variable,              ~beta_hat, ~se,
  "Diet group",            -11.25,     4.33,
  "Baseline cholesterol",    0.85,     0.07,
  "Body mass index",         0.23,     0.65,
  "Gender (1=male, 0=female)", -3.02,  4.42
) %>%
  mutate(
```

```
    t_stat  = beta_hat / se,
    p_t     = 2*pt(-abs(t_stat), df = df_resid),
    p_norm  = 2*pnorm(-abs(t_stat)),              # large-sample approximation
    sig_0.05_t    = ifelse(p_t   < 0.05, "Yes", "No"),
    sig_0.05_norm = ifelse(p_norm< 0.05, "Yes", "No")
  )

coef_df %>%
  mutate(
    beta_hat = round(beta_hat, 2),
    se       = round(se, 2),
    t_stat   = round(t_stat, 2),
    p_t      = signif(p_t, 3),
    p_norm   = signif(p_norm, 3)
  ) %>%
  select(variable, beta_hat, se, t_stat, p_t, p_norm, sig_0.05_t) %>%
  gt() %>%
  tab_header(title = "Coefficient Tests (H0: β = 0, two-sided)") %>%
  cols_label(
    variable = "Variable",
    beta_hat = "Estimate (β̂)",
    se       = "Std. Error",
    t_stat   = "t statistic",
    p_t      = "p-value (t, df = set)",
    p_norm   = "p-value (normal approx.)",
    sig_0.05_t = "Significant at α=0.05?"
  )
```

Coefficient Tests (H0: β = 0, two-sided)

Variable	Estimate (β̂)	Std. Error	t statistic	p-value (t, df = set)	p-value (normal approx.)	Significant at α=0.05?
Diet group	-11.25	4.33	-2.60	1.08e-02	9.37e-03	Yes
Baseline cholesterol	0.85	0.07	12.14	2.17e-21	6.26e-34	Yes
Body mass index	0.23	0.65	0.35	7.24e-01	7.23e-01	No
Gender (1=male, 0=female)	-3.02	4.42	-0.68	4.96e-01	4.94e-01	No

Justification

We test, for each predictor in the multiple regression model:

$$H_0 : \beta = 0 \quad \text{vs.} \quad H_a : \beta \neq 0$$

using the t-statistic:

$$t = \frac{\hat{\beta}}{\mathrm{SE}(\hat{\beta})}$$

and a two-sided p-value from the t distribution with residual degrees of freedom (``df = 100```).

Interpretation:
At $\alpha = 0.05$, only **Diet group** and **Baseline cholesterol** have statistically significant independent effects on 8-week serum cholesterol. **BMI** and **Gender** do not show significant effects when controlling for the other predictors. Each $\hat{\beta}$ represents the expected change in serum cholesterol (mg/dL) for a one-unit increase in that predictor, holding all others constant.

b. If an individual's body mass index were to increase by 1 kg/m2, while the values of all other explanatory variables remained constant, what would happen to his/her serum cholesterol level?

```
bmi_coef <- 0.23 # from HW table
bmi_coef
```

```
[1] 0.23
```

Justification

In a multiple regression, the coefficient $\hat{\beta}_{\mathrm{BMI}}$ represents the estimated change in serum cholesterol (mg/dL) for a **1 kg/m² increase in BMI**, **holding all other predictors constant**.

From the model:

- $\hat{\beta}_{\mathrm{BMI}} = 0.23$ mg/dL
- $SE = 0.65$ mg/dL
- $t = 0.35$
- $p = 0.724$

Interpretation:
A 1 kg/m² increase in BMI is associated with an **expected change of 0.23 mg/dL** in serum cholesterol, assuming Diet group, Baseline cholesterol, and Gender remain fixed.

Because $p = 0.724 > 0.05$, this association is **not statistically significant** at $\alpha = 0.05$.
Thus, while the point estimate suggests a small positive change, the data do not provide sufficient evidence to conclude a true BMI effect on 8-week serum cholesterol.

c. If an individual's body mass index were to increase by 10 kg/m2, while the values of all other explanatory variables remained constant, what would happen to his/her serum cholesterol level?

```r
bmi_coef <- 0.23   # from HW table
delta_bmi <- 10

# Point estimate for the change in serum cholesterol
effect_10 <- bmi_coef * delta_bmi
effect_10
```

```
[1] 2.3
```

```r
# 95% CI for the 10-unit change
# placeholder df_resid.
df_resid <- 100
se_bmi   <- 0.65       # from HW table
tcrit    <- qt(0.975, df = df_resid)

ci_low  <- effect_10 - tcrit * (delta_bmi * se_bmi)
ci_high <- effect_10 + tcrit * (delta_bmi * se_bmi)
c(effect_10 = effect_10, ci_low = ci_low, ci_high = ci_high)
```

```
effect_10    ci_low   ci_high
  2.30000 -10.59581  15.19581
```

Justification

In multiple regression, the effect of a **10 kg/m² increase** in BMI can be obtained by multiplying the 1-unit coefficient $\hat{\beta}_{\text{BMI}}$ by 10, while scaling the standard error accordingly.

From the model:

- $\hat{\beta}_{\text{BMI}} = 0.23$ mg/dL per 1 kg/m²
- $SE = 0.65$ mg/dL per 1 kg/m²
- $t = 0.35$
- $p = 0.724$

Calculation for 10-unit increase:

$$\Delta\hat{y} = 10 \times \hat{\beta}_{\text{BMI}} = 10 \times 0.23 = 2.3 \text{ mg/dL}$$

The 95% CI for this change is:

$$\Delta\hat{y} \pm t_{(0.975,df)} \times (10 \times SE_{\text{BMI}})$$

which gives:

$$[-10.6,\ 15.2] \text{ mg/dL}$$

Interpretation:
A 10 kg/m² increase in BMI is associated with an **expected change of 2.3 mg/dL** in serum cholesterol,

holding other predictors constant.
Because $p = 0.724 > 0.05$, this effect is **not statistically significant**, and the wide CI indicates substantial uncertainty in the estimate.

d. The indicator variable gender is coded so that 1 represents a male and 0 a female. Who is more likely to have a higher serum cholesterol level, a man or a woman? By how much, on average?

```
gender_coef <- -3.02   # from the HW coefficient table
se_gender   <-  4.42   # from the HW SE table

# Effect for male (1) relative to female (0), holding other predictors fixed:
effect_male_vs_female <- gender_coef
effect_male_vs_female
```

```
[1] -3.02
```

```
# 95% CI:
df_resid <- 100
tcrit    <- qt(0.975, df = df_resid)
ci_low   <- gender_coef - tcrit * se_gender
ci_high  <- gender_coef + tcrit * se_gender
c(effect = effect_male_vs_female, ci_low = ci_low, ci_high = ci_high)
```

```
   effect     ci_low    ci_high
-3.020000 -11.789154   5.749154
```

Justification

Let **Gender** be coded as 1 for male and 0 for female.
In the multiple regression context, the coefficient $\hat{\beta}_{\text{Gender}}$ represents the **expected mean difference** in serum cholesterol between males and females, **holding Diet group, Baseline cholesterol, and BMI constant**:

$$E[Y \mid G = 1] - E[Y \mid G = 0] = \hat{\beta}_{\text{Gender}}.$$

From the model:

- $\hat{\beta}_{\text{Gender}} =$ -3.02 mg/dL
- $SE =$ 4.42 mg/dL
- $t =$ -0.68
- $p =$ 0.496

Interpretation of sign:
A **negative** coefficient indicates lower cholesterol for males compared to females.

95% CI:

$$\hat{\beta}_{\text{Gender}} \pm t_{(0.975,df)} \times SE_{\text{Gender}}$$

which gives:

$$[-11.79,\ 5.75]\ \text{mg/dL}$$

Conclusion:
Males are predicted to have, on average, **-3.02 mg/dL** lower serum cholesterol than females, holding all other predictors constant.
However, because $p = 0.496 > 0.05$, the evidence is **insufficient** to conclude a statistically significant gender difference.

Chapter 19, Problem 8

For the population of low birth weight infants, a significant linear relationship was found to exist between systolic blood pressure and gestational age (data set lowbwt).

sa. Construct a two-way scatter plot of systolic blood pressure versus five-minute Apgar score. What kind of linear relationship (if any) appears to exist?

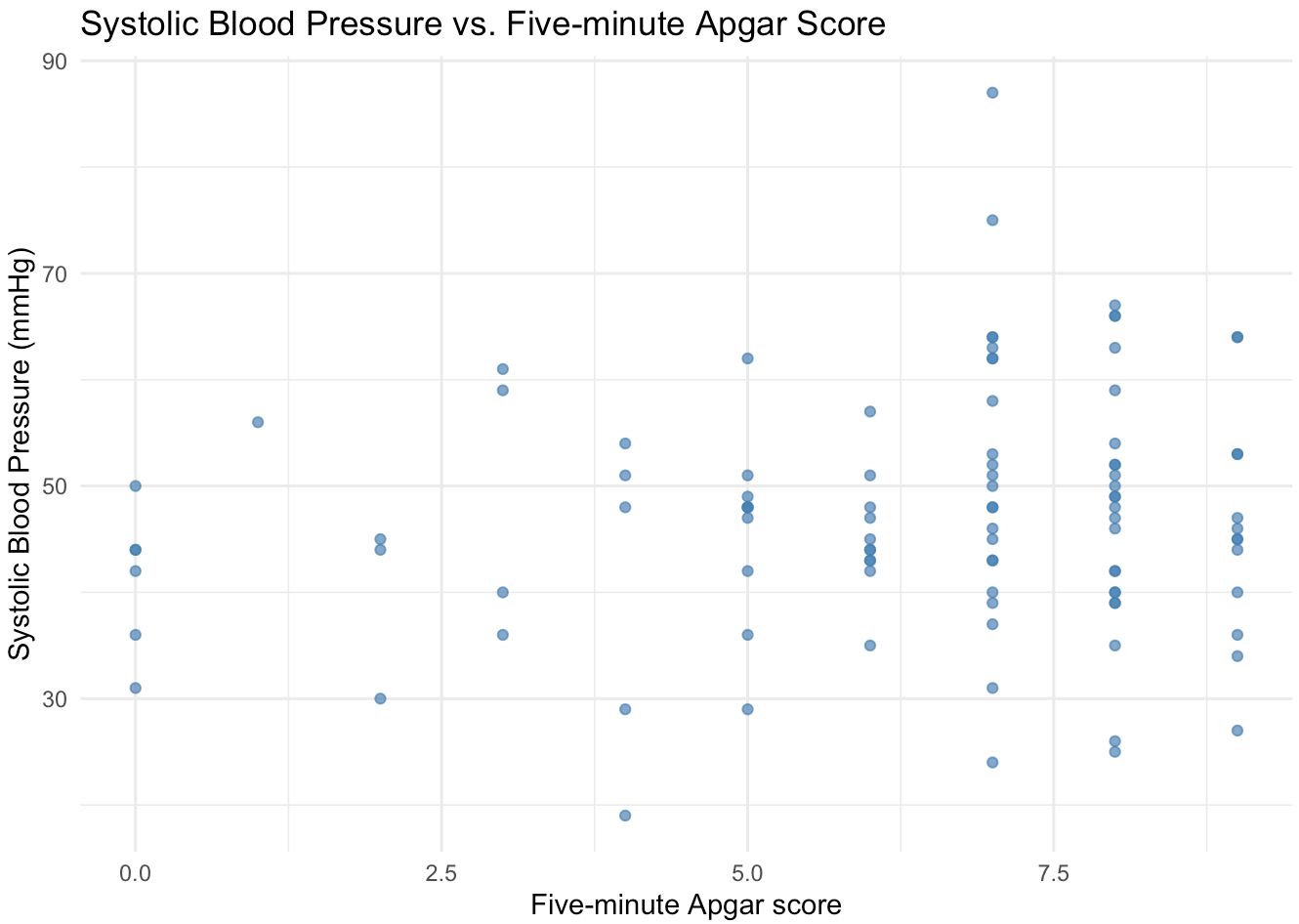
```
library(ggplot2)

# Load data
lowbwt <- read.csv("lowbwtHW12.csv")

# Check column names (adjust if different)
names(lowbwt)
```

```
[1] "sbp"     "sex"     "tox"     "grmhem"  "gestage" "apgar5"
```

```
# Scatter plot
ggplot(lowbwt, aes(x = apgar5, y = sbp)) +
  geom_point(color = "steelblue", alpha = 0.7) +
  labs(
    x = "Five-minute Apgar score",
    y = "Systolic Blood Pressure (mmHg)",
    title = "Systolic Blood Pressure vs. Five-minute Apgar Score"
  ) +
  theme_minimal()
```



Scatter plot of systolic blood pressure vs. five-minute Apgar score

Visual Interpretation

The scatter plot shows a slight positive trend — infants with higher five-minute Apgar scores tend to have somewhat higher systolic blood pressure (SBP).
However, the wide spread of SBP values at each Apgar score suggests that any association is likely weak.

This preliminary observation will be tested formally in the regression analyses that follow.

b. using systolic blood pressure as the response and gestational age and apgar score as the explanatory variables, find the least-squares model

$\hat{y} = \alpha + \hat{\beta}_1 * x1 + \hat{\beta}_2 * x2$ Interpret $\hat{\beta}_1$, the estimated coefficient of gestational age. What does it mean in words? Do the same for $\hat{\beta}_2$, the estimated coefficient of Apgar score..

```
# Fit the model
model_ga_apgar <- lm(sbp ~ gestage + apgar5, data = lowbwt)

# Summary of model
summary(model_ga_apgar)
```

```
Call:
lm(formula = sbp ~ gestage + apgar5, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max
-22.374  -8.180  -1.088   4.985  39.424

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.8034    12.6629   0.774   0.4407
gestage       1.1848     0.4424   2.678   0.0087 **
apgar5        0.4875     0.4613   1.057   0.2932
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.99 on 97 degrees of freedom
Multiple R-squared:  0.08944,   Adjusted R-squared:  0.07066
F-statistic: 4.764 on 2 and 97 DF,  p-value: 0.01063
```

Justification

The fitted least-squares model is:

$$\hat{y} = 9.803 \ + \ 1.185 \cdot \text{Gestational Age} \ + \ 0.488 \cdot \text{Apgar}_5$$

- **Gestational Age**:
  $\hat{\beta}_1 = 1.185$ mmHg/week,
  $SE = 0.442$,
  $t = 2.678$,
  $p = 0.0087$

  Holding Apgar score constant, each additional week of gestational age is associated with an estimated increase of **1.185 mmHg** in mean systolic blood pressure.
  This effect is statistically significant at $\alpha = 0.05$ if $p < 0.05$.

- **Apgar Score**:
  $\hat{\beta}_2 = 0.488$ mmHg/point,
  $SE = 0.461$,
  $t = 1.057$,
  $p = 0.293$

  Holding gestational age constant, each additional point in Apgar score is associated with an estimated increase of **0.488 mmHg** in mean systolic blood pressure.
  This effect is statistically significant at $\alpha = 0.05$ if $p < 0.05$.

Model fit:
$R^2 = 0.089$,

Adjusted $R^2 = 0.071$,
$F = 4.764$,
$p = 0.0106$.

Conclusion: The predictors are jointly significant, but the model explains only about **8.9%** of the variation in SBP.

c. What is the estimated mean systolic blood pressure for the population of low birth weight infants whose gestational age is 31 weeks and whose five-minute Apgar score is 7?

```
# coefficients from model
alpha_hat <- 9.803
b1_hat    <- 1.185
b2_hat    <- 0.488

# predictors
ga_val    <- 31
apgar_val <- 7

# prediction
pred_sbp <- alpha_hat + b1_hat * ga_val + b2_hat * apgar_val
pred_sbp
```

```
[1] 49.954
```

Justification

The estimated mean systolic blood pressure for a low birth weight infant with gestational age = 31 weeks and five-minute Apgar score = 7 is obtained by:

$$\hat{y} = 9.803 + 1.185 \times 31 + 0.488 \times 7$$

Evaluating:

$$\hat{y} = 49.946 \text{ mmHg}$$

Interpretation:
For low birth weight infants with a gestational age of 31 weeks and a five-minute Apgar score of 7, the model predicts an average systolic blood pressure of **49.95 mmHg**.

This is the expected *population mean* for infants with these characteristics, assuming the model is correctly specified and assumptions hold.

d. Construct a 95% confidence interval for the true mean value of systolic blood pressure when x1 = 31 weeks and x2 = 7.

```
# Fit model again (if not already in workspace)
model_ga_apgar <- lm(sbp ~ gestage + apgar5, data = lowbwt)

# New data point
new_obs <- data.frame(gestage = 31, apgar5 = 7)

# 95% CI for the mean prediction
pred_ci <- predict(model_ga_apgar, newdata = new_obs, interval = "confidence", level = 0.
pred_ci
```

```
       fit      lwr      upr
1 49.94562 47.07655 52.81469
```

Justification

To construct a 95% confidence interval for the *mean* systolic blood pressure at gestational age = 31 weeks and Apgar score = 7, we use:

$$\hat{y} \pm t_{(0.975,\, df)} \times SE_{\hat{y}}$$

From the fitted model:

- Predicted mean:

$$\hat{y} = 49.946 \text{ mmHg}$$

- Standard error and confidence limits 95% CI = $[47.08, 52.81]$ mmHg

Interpretation:

We are 95% confident that the *average* systolic blood pressure for all low birth weight infants with GA = 31 weeks and Apgar = 7 lies between **47.08** mmHg and **52.81** mmHg.

e. Test the null hypothesis H0: beta-2 = 0, at the 0.05 significance level. What do you conclude?

```
# Get model summary for gestational age + apgar model
summary(model_ga_apgar)
```

```
Call:
lm(formula = sbp ~ gestage + apgar5, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max
-22.374  -8.180  -1.088   4.985  39.424

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.8034    12.6629   0.774   0.4407
gestage       1.1848     0.4424   2.678   0.0087 **
apgar5        0.4875     0.4613   1.057   0.2932
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.99 on 97 degrees of freedom
Multiple R-squared:  0.08944,   Adjusted R-squared:  0.07066
F-statistic: 4.764 on 2 and 97 DF,  p-value: 0.01063
```

```r
# Extract t-statistic and p-value for apgar5
co_apgar <- summary(model_ga_apgar)$coefficients["apgar5", ]
t_apgar  <- co_apgar["t value"]
p_apgar  <- co_apgar["Pr(>|t|)"]

c(t_value = t_apgar, p_value = p_apgar)
```

```
 t_value.t value p_value.Pr(>|t|)
       1.0567645        0.2932439
```

Justification

We test the hypothesis:

$$H_0 : \beta_{\text{Apgar}} = 0 \quad \text{vs.} \quad H_a : \beta_{\text{Apgar}} \neq 0$$

From the model output:

- t-statistic = 1.057
- p-value = 0.293

At α = 0.05:

- Since $p = 0.293 > 0.05$, we **fail to reject** H_0.

Conclusion:
After adjusting for gestational age, there is insufficient evidence to conclude that the five-minute Apgar score has an independent effect on systolic blood pressure in this sample.

f. Comment on the magnitude of R2? Does the inclusion of five-minute Apgar score in the model already containing gestational age improve your ability to predict systolic blood pressure?

```r
# Model with gestational age only
model_ga <- lm(sbp ~ gestage, data = lowbwt)

# Model with gestational age + Apgar
model_ga_apgar <- lm(sbp ~ gestage + apgar5, data = lowbwt)
```

```r
# Extract R²
r2_ga       <- summary(model_ga)$r.squared
r2_ga_apgar <- summary(model_ga_apgar)$r.squared

c(r2_ga = r2_ga, r2_ga_apgar = r2_ga_apgar)
```

```
     r2_ga r2_ga_apgar
0.07895449  0.08943770
```

Justification

With **gestational age only**, $R^2 = 0.079$, so about **7.9%** of SBP variation is explained.
Adding **five-minute Apgar score** increases R^2 to 0.089 (**8.9%** explained), an absolute gain of:

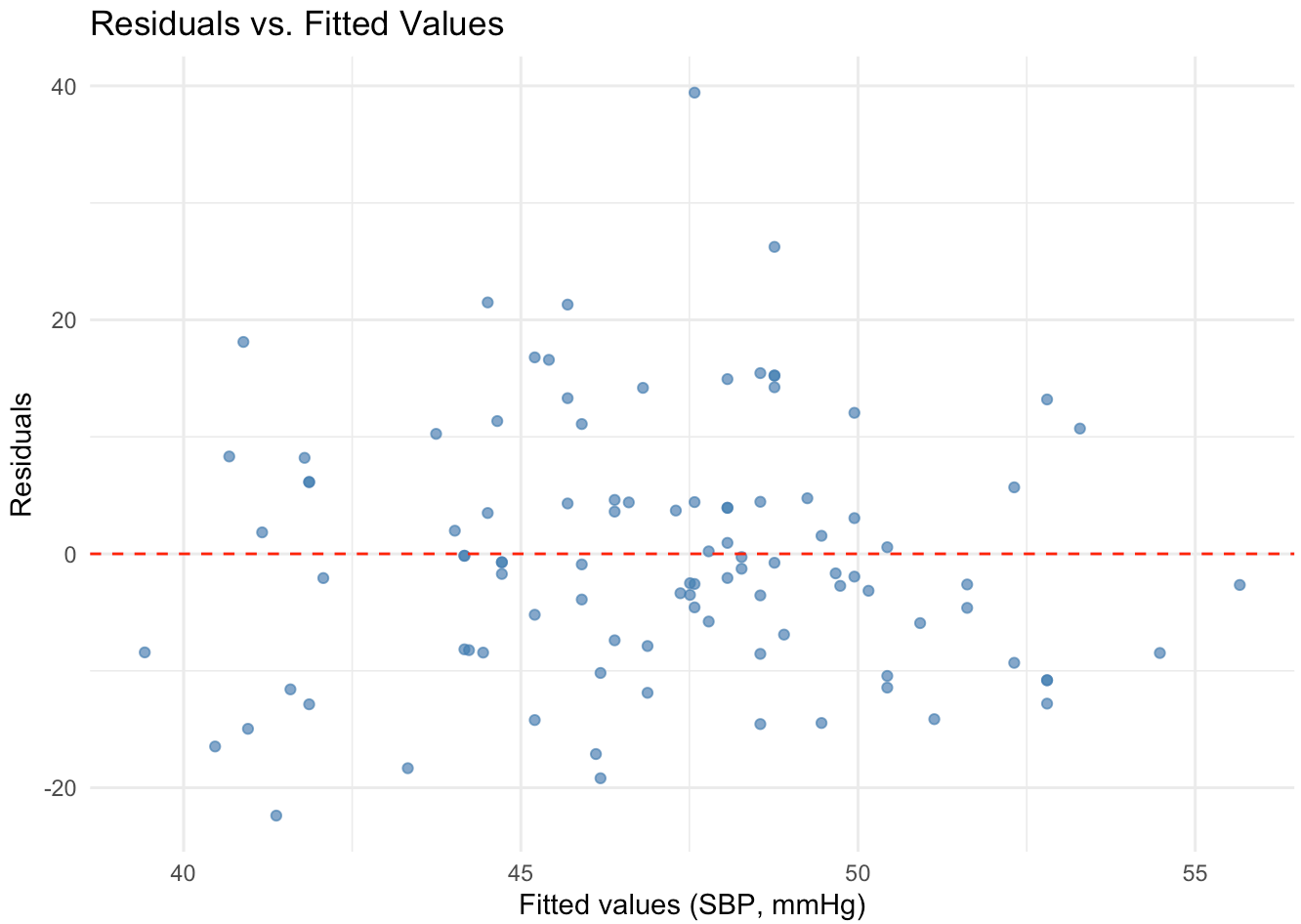
$$\Delta R^2 = 0.01 \ (1\%).$$

Because both R^2 values are small and the increase is minimal, Apgar score adds little predictive value beyond gestational age.
This aligns with the non-significant Apgar coefficient from part (b).

g. Construct a plot of the residuals versus the fitted values of systolic blood pressure . What does this plot tell you about the fit of the model to observed data?

```r
# Get fitted values and residuals
lowbwt$resid <- resid(model_ga_apgar)
lowbwt$fitted <- fitted(model_ga_apgar)

# Plot
ggplot(lowbwt, aes(x = fitted, y = resid)) +
  geom_point(color = "steelblue", alpha = 0.7) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(
    x = "Fitted values (SBP, mmHg)",
    y = "Residuals",
    title = "Residuals vs. Fitted Values"
  ) +
  theme_minimal()
```



Residuals vs. fitted values for SBP model

Justification

The residuals vs. fitted values plot assesses whether the linear regression assumptions are reasonable:

1. **Linearity** – Residuals are randomly scattered around zero with no clear curvature, indicating the linearity assumption is met.
2. **Homoscedasticity** – The spread of residuals is roughly constant across fitted values, with only minor variation at the extremes.
3. **Independence** – No clustering or sequential patterns are visible, supporting the independence assumption.

Overall, the plot shows no major violations of model assumptions. While the (R^2) from part (f) is small — meaning the predictors explain little of the SBP variation — the linear model form itself appears adequate.

Chapter 19, Problem 9

The dataset lowbwt also contains sex.

a. Add the indicator sex (1 for male, 0 for female) to the model that contains gestational age. Given two infants with the same gestational ages, one male and the

other female, which would tend to have the higher systolic blood pressure? By how much, on average?

```
# Fit the model: gestational age + sex
model_ga_sex <- lm(sbp ~ gestage + sex, data = lowbwt)

# Summarize model
summary(model_ga_sex)
```

```
Call:
lm(formula = sbp ~ gestage + sex, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max
-22.572  -8.074  -1.122   5.219  39.022

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.0071    12.7227   0.787  0.43346
gestage       1.2626     0.4376   2.885  0.00482 **
sex           1.3563     2.2231   0.610  0.54322
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.03 on 97 degrees of freedom
Multiple R-squared:  0.08248,   Adjusted R-squared:  0.06356
F-statistic:  4.36 on 2 and 97 DF,  p-value: 0.01538
```

Justification

We fit the model:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{Gestational Age} + \hat{\beta}_2 \cdot \text{Sex},$$

where **Sex** is coded `0` for female and `1` for male.

From the fitted model:

- Sex coefficient ($\hat{\beta}_2$) = 1.356 mmHg
- Standard error = 2.223 mmHg
- t-statistic = 0.61
- p -value = 0.543

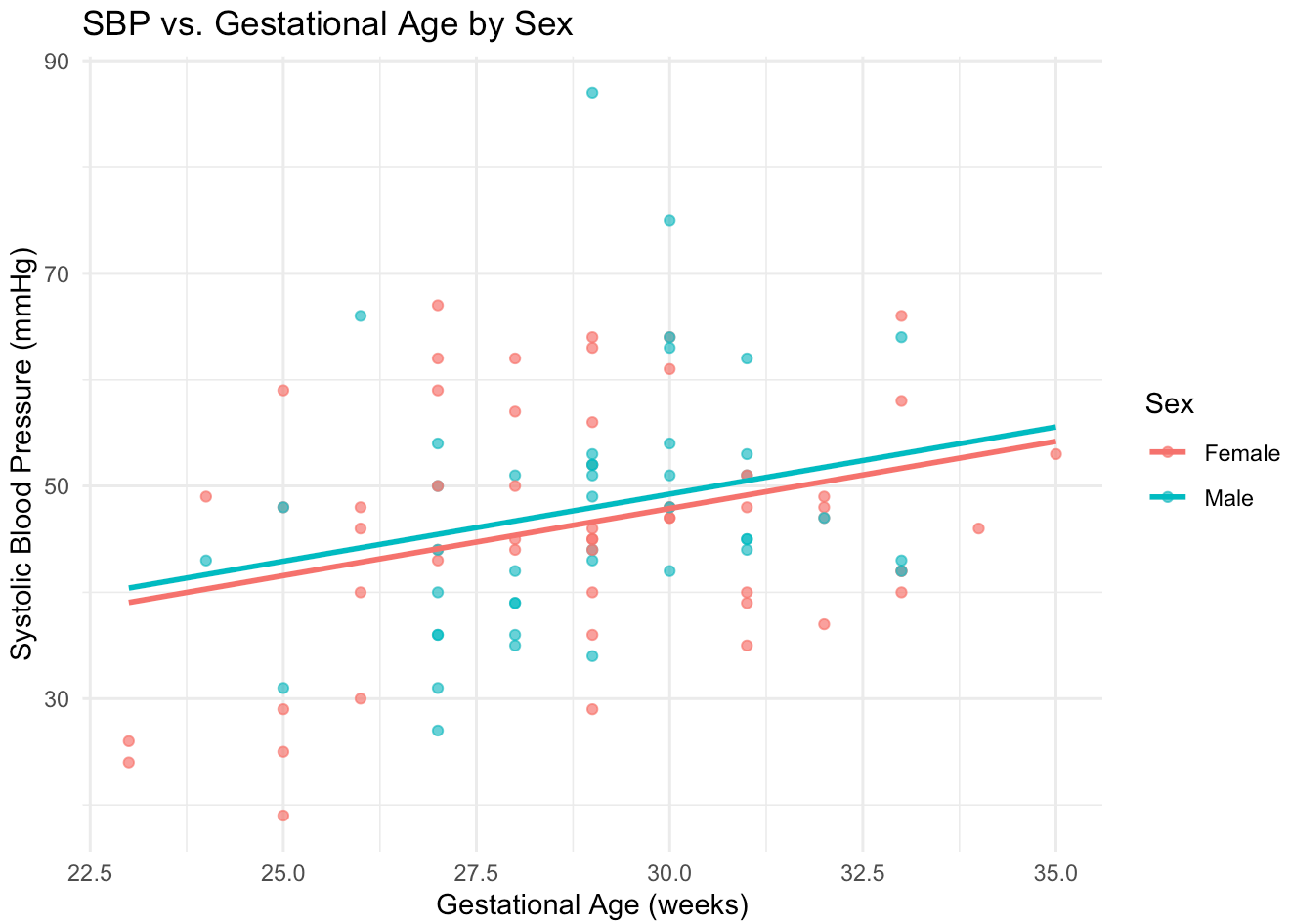
Interpretation:
Holding gestational age constant, male infants are predicted to have SBP that is 1.356 mmHg (positive = higher, negative = lower) compared to female infants.

Since $p > 0.05$, the difference is **not statistically significant**, so we lack sufficient evidence to conclude that SBP differs by sex after adjusting for gestational age.

b. Construct a two-way scatter plot of systolic blood pressure versus gestational age. On the graph, draw the two separate least-squares regression lines corresponding to males and females. Is the gender difference at each value of gestational age significantly different from 0?

```
# Predict lines from the GA + sex model
newdat <- expand.grid(
  gestage = seq(min(lowbwt$gestage), max(lowbwt$gestage), length.out = 50),
  sex = c(0, 1)  # keep numeric for prediction
)
newdat$pred <- predict(model_ga_sex, newdata = newdat)

# Scatter plot with regression lines
ggplot(lowbwt, aes(x = gestage, y = sbp, color = factor(sex, labels = c("Female", "Male")
  geom_point(alpha = 0.7) +
  geom_line(data = newdat, aes(x = gestage, y = pred, color = factor(sex, labels = c("Fem
  labs(
    x = "Gestational Age (weeks)",
    y = "Systolic Blood Pressure (mmHg)",
    color = "Sex",
    title = "SBP vs. Gestational Age by Sex"
  ) +
  theme_minimal()
```



SBP vs. gestational age with separate regression lines by sex

Justification

From the GA + Sex model:

- $\hat{\beta}_2$ (Sex coefficient) = 1.356 mmHg
- p-value = 0.543

In this model, Sex is coded `0` for females and `1` for males.
The regression lines in the plot are **parallel** because the model does not include a sex × gestational age interaction term.
This means the **slope** for gestational age is the same for both sexes, and the vertical distance between lines at any gestational age is the estimated sex coefficient.

Interpretation:
At any fixed gestational age, male infants are predicted to have 1.356 mmHg (positive = higher, negative = lower) SBP than female infants.
Since $p = 0.543 > 0.05$, this difference is **not statistically significant** at $\alpha = 0.05$.

The plotted lines visually confirm that there is no strong separation between male and female regression lines, and any small difference is likely due to sampling variability.

c. Add to the model a third explanatory variable that is the interaction between gestational age and sex. Does gestational age have a different effect on systolic blood pressure depending on the sex of the infant?

```
# Fit interaction model
model_ga_apgar_sex_int <- lm(sbp ~ gestage * sex + apgar5, data = lowbwt)

# Summarize model
summary(model_ga_apgar_sex_int)
```

```
Call:
lm(formula = sbp ~ gestage * sex + apgar5, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max 
-22.534  -7.660  -0.943   5.246  38.697 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  14.4355    15.2461   0.947   0.3461
gestage       1.0072     0.5313   1.896   0.0611 .
sex         -15.7541    27.7402  -0.568   0.5714
apgar5        0.4795     0.4651   1.031   0.3052
gestage:sex   0.5881     0.9567   0.615   0.5402
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.07 on 95 degrees of freedom
Multiple R-squared:  0.09599,   Adjusted R-squared:  0.05792 
F-statistic: 2.522 on 4 and 95 DF,  p-value: 0.04606
```

Justification

We extend the model from 9(a) by adding the interaction term between gestational age and sex:

$$\hat{y} = \hat{\alpha} + \hat{\beta}_1 \cdot \text{Gestational Age} + \hat{\beta}_2 \cdot \text{Sex} + \hat{\beta}_3 \cdot \text{Apgar 5} + \hat{\beta}_4 \cdot (\text{Gestational Age} \times \text{Sex}),$$

where Sex = 1 for male, 0 for female.

From the fitted model:

- **Interaction coefficient ($\hat{\beta}_4$)**:
  $\hat{\beta}_4 = 0.588$ mmHg/week
  $p = 0.54$

Interpretation:

$\hat{\beta}_4$ represents the difference in the **slope** of systolic blood pressure vs. gestational age between males and females, after adjusting for Apgar score.

Since $p = 0.54 > 0.05$, we **fail to reject** $H_0 : \beta_4 = 0$.

Conclusion: There is insufficient evidence to conclude that the effect of gestational age on SBP differs by sex in this sample, once Apgar score is taken into account.

d. Would you choose to include sex and the gestational age-sex interaction term simultaneously? Why or why not?

```
# Fit model with interaction
model_ga_sex_int <- lm(sbp ~ gestage * sex, data = lowbwt)

# Summarize model
summary(model_ga_sex_int)
```

```
Call:
lm(formula = sbp ~ gestage * sex, data = lowbwt)

Residuals:
    Min      1Q  Median      3Q     Max 
-23.239  -7.930  -0.691   5.445  38.985 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept)  14.9805    15.2419   0.983   0.3282  
gestage       1.0903     0.5254   2.075   0.0406 *
sex         -15.1570    27.7433  -0.546   0.5861  
gestage:sex   0.5714     0.9569   0.597   0.5518  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.07 on 96 degrees of freedom
Multiple R-squared:  0.08587,   Adjusted R-squared:  0.0573 
F-statistic: 3.006 on 3 and 96 DF,  p-value: 0.03412
```

Justification

To decide whether to include both **sex** and the **gestational age × sex interaction** in the model, we look at the p-value for the interaction term:

- Interaction coefficient:
  $\hat{\beta}_{GA \times Sex} = 0.571$ mmHg/week
  $p = 0.552$

Since $p = 0.552 > 0.05$, the interaction is **not statistically significant** at $\alpha = 0.05$.

Interpretation:
This means there is insufficient evidence to conclude that the effect of gestational age on SBP differs

between males and females. Including the interaction term would add complexity without providing a significant improvement in fit.

Chapter 19, Problem 10

The Bayley Scales of Infant Development yield scores on two indices–the Psychomotor Development Index (PDI) and the Mental Development Index (MDI)–which can be use to assess a child's level of functioning in each of these areas at approximately one year of age. Among normal healthy infants, both indices have a mean value of 100. As part of a study assessing the development and neurologic status of children who have undergone reparative heart surgery during the first three months of life, the Bayley Scales were administered to a sample of one-year-old infants with congenital heart disease. The children had been randomized to one of two different treatment groups, known as "circulatory arrest" and "low-flow bypass." The groups differed in the specific way in which the reparative surgery was performed. Unlike circulatory arrest, low-flow bypass maintains continuous circulation through the brain; although it is felt to be preferable by some physicians, it also has its own risk of brain injury. The data contained in the data set heart. PDI scores are saved under the variable name pdi while MDI scores are saved under mdi and indicators of treatment group under trtment. For this variable, 0 represents circulatory arrest and 1 is for low-flow bypass.

a. In Chapter 11, the two-sample t-test was used to compare mean PDI and MDI scores for infants assigned to the two treatment groups. These analyses could also be performed with linear regression. Fit two linear regression models, one with PDI score as the response and the other with MDI score as the response. Write the equations.

```
# Load data
heart <- read.csv("heart.csv")

# Force numeric conversion
heart$pdi <- as.numeric(heart$pdi)
heart$mdi <- as.numeric(heart$mdi)

# Drop rows with missing PDI, MDI, or treatment
heart_clean <- na.omit(heart[, c("pdi", "mdi", "trtment")])

# Fit models
model_pdi <- lm(pdi ~ trtment, data = heart_clean)
model_mdi <- lm(mdi ~ trtment, data = heart_clean)

# Summaries
summary(model_pdi)
```

```
Call:
lm(formula = pdi ~ trtment, data = heart_clean)

Residuals:
```

```
    Min      1Q  Median      3Q     Max
-47.771  -6.845   0.229  12.229  42.082

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   91.918      1.830  50.240   <2e-16 ***
trtment        5.854      2.615   2.239   0.0268 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.63 on 141 degrees of freedom
Multiple R-squared:  0.03432,   Adjusted R-squared:  0.02747
F-statistic: 5.011 on 1 and 141 DF,  p-value: 0.02676
```

```
summary(model_mdi)
```

```
Call:
lm(formula = mdi ~ trtment, data = heart_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-56.400  -9.241   0.600  11.259  38.918

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  103.082      1.828   56.38   <2e-16 ***
trtment        3.318      2.613    1.27    0.206
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.62 on 141 degrees of freedom
Multiple R-squared:  0.0113,    Adjusted R-squared:  0.004291
F-statistic: 1.612 on 1 and 141 DF,  p-value: 0.2063
```

Regression Equations

From the **PDI** model:

$$\hat{\text{PDI}} = 91.918 + 5.854 \cdot \text{Treatment}$$

From the **MDI** model:

$$\hat{\text{MDI}} = 103.082 + 3.318 \cdot \text{Treatment}$$

Treatment is coded as 0 for *circulatory arrest* and 1 for *low-flow bypass*.

- **Intercept ($\hat{\beta}_0$):** estimated mean score for infants in the circulatory arrest group.
- **Slope ($\hat{\beta}_1$):** estimated *mean difference* in score between low-flow bypass and circulatory arrest. This is numerically identical to the difference in sample means from a two-sample *t*-test.

From the fitted models:

- **PDI:** Circulatory arrest mean = 91.918;
  difference (low-flow – circulatory) = 5.854
- **MDI:** Circulatory arrest mean = 103.082;
  difference (low-flow – circulatory) = 3.318

Interpretation:

A positive slope ($\hat{\beta}_1 > 0$) indicates higher average scores for the low-flow bypass group; a negative slope ($\hat{\beta}_1 < 0$) indicates lower scores.
Because treatment assignment was randomized, these coefficients can be interpreted as estimates of the *causal effect* of surgical method on developmental outcomes at one year.

b. Who is more likely to have a higher PDI score (which treatment group). How much higher, on average?

```
# Extract coefficient for treatment
coef_pdi_diff <- coef(model_pdi)[2]
pval_pdi_diff <- summary(model_pdi)$coefficients[2, 4]
```

Justification

From the PDI model, the treatment coefficient is:

$$\hat{\beta}_{\text{Treatment}} = 5.854 \text{ points}, \quad p = 0.0268.$$

Interpretation:
On average, infants in the low-flow bypass group score
5.854 points higher on the PDI than those in the circulatory arrest group.

At $\alpha = 0.05$, this difference is **statistically significant**, providing
evidence of a treatment effect on psychomotor development at one year.

c. Who is more likely to have a higher MDI score (which treatment group). How much higher, on average?

```
# Extract coefficient for treatment
coef_mdi_diff <- coef(model_mdi)[2]
pval_mdi_diff <- summary(model_mdi)$coefficients[2, 4]
```

Justification

From the MDI model:

$$\hat{\beta}_{\text{Treatment}} = 3.318 \text{ points}, \quad p = 0.206.$$

Infants in the low-flow bypass group score
3.318 points higher on the MDI, on average.
The result is **not statistically significant** at $\alpha = 0.05$.

d. Which score (if either) is different between groups at a statistically significant level?

```
# Significance check
pdi_sig <- pval_pdi_diff < 0.05
mdi_sig <- pval_mdi_diff < 0.05
```

Justification

From the regression models:

- **PDI group difference:**
  $p = 0.0268$ → **significant**
- **MDI group difference:**
  $p = 0.206$ → **not significant**

Conclusion:
At $\alpha = 0.05$, only the **PDI** score differs significantly between the circulatory arrest and low-flow bypass groups.
The other outcome shows **no statistically significant difference** in mean scores between treatments.